



Solution Set

Question 1

Integrate the right side and apply

$$\begin{aligned}y(0) &= 0 \\y(x) &= \int (x^2 + e^{3x} + 1) dx = \frac{x^3}{3} + \frac{e^{3x}}{3} + x + C \\0 &= \frac{1}{3} + C \Rightarrow C = -\frac{1}{3} \\y(x) &= \frac{x^3}{3} + x + \frac{e^{3x} - 1}{3}\end{aligned}$$

Question 2

The constant function

$$y(x) = 1$$

satisfies the equation because

$$y' = 0, \quad \sqrt{1 - y^2} = 0$$

and it satisfies

$$y(0) = 1$$

So the IVP solution is

$$y(x) = 1$$

Question 3

Continuous compounding means the instantaneous growth rate is proportional to the current account value.

$$\begin{aligned}\frac{dA}{dt} &= \rho A, \quad \rho = \frac{r}{100} \\ \frac{dA}{A} &= \rho dt \Rightarrow A(t) = A(0)e^{\rho t}\end{aligned}$$

Question 4

Natural growth has

$$N(t) = N_0 e^{kt}$$

Tenfold in 10 hours gives

$$10 = e^{10k} \Rightarrow k = \frac{\log 10}{10}$$

Doubling time solves

$$2 = e^{kt_d} \Rightarrow t_d = \frac{\log 2}{k} = 10 \frac{\log 2}{\log 10} \approx 3.01 \text{ hours}$$

Question 5

The learning curve is

$$\frac{dA}{dt} = k(M - A)$$

Solving:

$$A(t) = M - (M - A_0)e^{-kt}$$

With

$$A_0 = 0, \quad M = 1, \quad A(100) = 0.5$$

$$0.5 = 1 - e^{-100k} \Rightarrow k = \frac{\log 2}{100}$$

For 75 percent:

$$0.75 = 1 - e^{-kt} \Rightarrow t = \frac{\log 4}{k} = 200$$

Additional study required is 100 hours.

Question 6

Rewrite as

$$y' + \frac{3}{x}y = 8x^4$$

The integrating factor is

$$\mu(x) = e^{\int 3/x \, dx} = x^3$$

Then

$$(x^3 y)' = 8x^7$$
$$x^3 y = x^8 + C \Rightarrow y = x^5 + Cx^{-3}$$

Question 7

Let

$$y = Ax + Bx^2 + 2x^3$$

Then

$$y' = A + 2Bx + 6x^2, \quad y'' = 2B + 12x$$

Substitute:

$$x^2 y'' - 2xy' + 2y = (2Bx^2 + 12x^3) - 2x(A + 2Bx + 6x^2) + 2(Ax + Bx^2 + 2x^3) = 4x^3$$

so the proposed solution satisfies the differential equation.